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Bickerstaff's brainstem encephalitis and Fisher syndrome form a continuous spectrum

Clinical analysis of 581 cases

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Abstract Whether Bickerstaff's brainstem encephalitis (BBE) is a distinct disease or a subtype of Fisher syndrome (FS) is unclear as there have been no clinical studies with sufficiently large numbers of patients with FS or BBE. Our aim was to clarify the nosological relationship. Medical records of patients suffering acute ophthalmoplegia and ataxia within four weeks of onset were reviewed. BBE was the diagnosis for patients with impaired consciousness, FS for those with clear consciousness and areflexia. Clinical features, neuroimages, and laboratory findings were analyzed. Patients were grouped as having BBE ($n = 53$), FS ($n = 466$), or as unclassified ($n = 62$). The BBE and FS groups had similar features; positive serum anti-GQ1b IgG antibody (68 % versus 83 %), antecedent *Campylobacter jejuni* infection (23 % versus 21 %), CSF albuminocytological dissociation

(46 % versus 76 %), brain MRI abnormality (11 % versus 2 %), and abnormal EEG findings (57 % versus 25 %). BBE ($n = 4$) and FS ($n = 28$) subgroups underwent detailed electrophysiological testing. Both groups frequently showed absent soleus H-reflexes, but normal sensory nerve conduction (75 % versus 74 %) and a 1-Hz power spectrum peak on postural body sway analysis (67 % versus 72 %). Common autoantibodies, antecedent infections, and MRI and neurophysiological results found in this large study offer conclusive evidence that Bickerstaff's brainstem encephalitis and Fisher syndrome form a continuous spectrum with variable CNS and PNS involvement.

Key words Anti-GQ1b antibody · Bickerstaff's brainstem encephalitis · *Campylobacter jejuni* · Fisher syndrome · Guillain-Barré syndrome

Introduction

Fisher [10] reported three cases of acute ophthalmoplegia, ataxia, and areflexia subsequent to upper respiratory infectious symptoms, and postulated that the disorder is a Guillain-Barré syndrome (GBS) variant because of CSF albuminocytological dissociation in one. Some patients with Fisher syndrome (FS) progress to GBS [22], proving Fisher's postulation. Bickerstaff [4] reported eight such cases and named the condition "brain-stem encephalitis",

a definitive syndrome. Seven of those eight patients had symmetrical ophthalmoplegia, ataxia, and impaired consciousness. The etiology of Bickerstaff's brainstem encephalitis (BBE) was speculated to be similar to that of GBS because prodromal upper respiratory infection and CSF albuminocytological dissociation were present. One of Bickerstaff's original patients had flaccid limb weakness. In a study that included 62 patients with BBE, 37 had limb weakness, considered the result of an overlapping axonal subtype of GBS [26]. These clinical findings indicate that BBE and GBS are closely related

and on a continuous spectrum, as Bickerstaff assumed [4].

One of the three patients originally described by Fisher [10] experienced drowsiness, and four of the eight patients described by Bickerstaff [4] had hypo- or areflexia. Subsequently, Bickerstaff's group reported 18 other patients who had "brainstem encephalitis and the syndrome of Miller Fisher" and argued for a central origin [1]. All 18 had ophthalmoplegia and ataxia. Eleven experienced drowsiness, and one was comatose. Tendon reflexes were absent in 11, normal in three, and brisk in four. Based on radiological (three patients) and pathological (one patient) changes in the brainstem, as well as abnormal EEG findings (12 patients), they insisted that the responsible lesion is localized in the CNS. Ropper [30], who criticized their report, considered six of the 18 cases typical FS, and the other 12 obscure brainstem lesions without peripheral polyneuropathy.

Serum IgG antibody to ganglioside GQ1b, reported as a possible FS marker [7], was also detected in BBE patients [35]. The immunological profile common to FS and BBE supports a common pathogenesis. A comprehensive, case-control study was made that identified serologically antecedent infectious agents in FS and GBS [16] but not in BBE. Because of the lack of studies of large numbers of patients with FS or BBE, it is unclear whether BBE is a distinct disease or FS subtype. For our study we therefore reviewed medical records of 581 patients who had acute ophthalmoplegia and ataxia but no significant limb weakness. We analyzed their clinical features and laboratory findings and obtained conclusive evidence that FS and BBE form a continuous spectrum with variable CNS and PNS involvement.

Methods

Patients and diagnostic criteria

Between 1999 and 2006, approximately 9200 patients were referred for serum anti-ganglioside antibody testing from hospitals throughout Japan, which excluded the BBE patients reported in our previous study [26]. The Ethical Committee at Dokkyo Medical University approved performance of this study. Clinical information was obtained from each primary physician. CSF albuminocytological dissociation was defined as a raised protein concentration (more than 45 mg/dl) associated with a count of 10 or fewer leukocytes/ μ l. CSF pleocytosis was defined as the increased count of 11 or more leukocytes/ μ l.

Based on Fisher's [10] and Bickerstaff's [4] findings, "progressive, relatively symmetric external ophthalmoplegia and ataxia by four weeks" are the clinical features necessary for a diagnosis of both BBE and FS. Whereas "hypo- or areflexia" and clear consciousness were required for the diagnosis of FS, "impaired consciousness" is for that of BBE. Hypo- or areflexia was not an exclusion criterion for the diagnosis of BBE because half of the original cases had hypo- or areflexia. In our study, "severe limb weakness" of 3 or less on the Medical Research Council (MRC) scale was an exclusion criterion for both conditions, although there was gross flaccid limb weakness in Bickerstaff's Case 4 [4]. In contrast, limb weakness of 4 on the MRC scale was an inclusion criterion for either condition because there was mild limb weakness in

Fisher's original Case 1 [10]. For the diagnosis of either condition, the following must be excluded: vascular disease involving the brainstem, Wernicke's encephalopathy, botulism, myasthenia gravis, brainstem tumor, pituitary apoplexy, acute disseminated encephalomyelitis (disorders described by Bickerstaff and Cloake, Fisher, Bickerstaff, and Al-Din et al.) [1, 4, 5, 10], multiple sclerosis, neuro-Behçet disease, vasculitis, lymphoma, and Creutzfeldt-Jakob disease (our addition). Based on GBS criteria [2], extensor plantar reflexes do not exclude an FS diagnosis if other clinical features are typical.

Anti-ganglioside and infectious serology

Serum IgG antibodies to GQ1b, GT1a, GM1b, GalNAc-GD1a, and GD1b (Fig. 1) were measured by an enzyme-linked immunosorbent assay [37]. Serum was considered positive for anti-ganglioside antibodies when the absorbance value was 0.1 or higher at a dilution 1:500.

Sera obtained from 34 patients with BBE, 73 with FS, and 44 with unclassified conditions (see Patient selection in Results) were available for infection serology tests. Evidence of recent infections by *Campylobacter jejuni*, *Haemophilus influenzae*, cytomegalovirus, Epstein-Barr virus, and *Mycoplasma pneumoniae* were assayed serologically [16]: Sera from 73 hospital control subjects were used. Frequency differences between groups were compared by the χ^2 or Fisher's exact (two-tailed) test. A *p* value of less than 0.05 was considered significant.

Nerve conduction study, H-reflex, somatosensory evoked potential, and postural body sway analyses

A subgroup of BBE patients (*n* = 4) or FS (*n* = 28), seen consecutively at Chiba University Hospital, underwent nerve conduction and H-reflex studies as well as somatosensory evoked potentials and postural body sway analyses. Motor nerve conduction studies, including F-wave ones, of the median, ulnar, peroneal, and tibial nerves were made by conventional procedures. Antidromic sensory nerve action potentials were recorded in the median, ulnar, superficial radial,

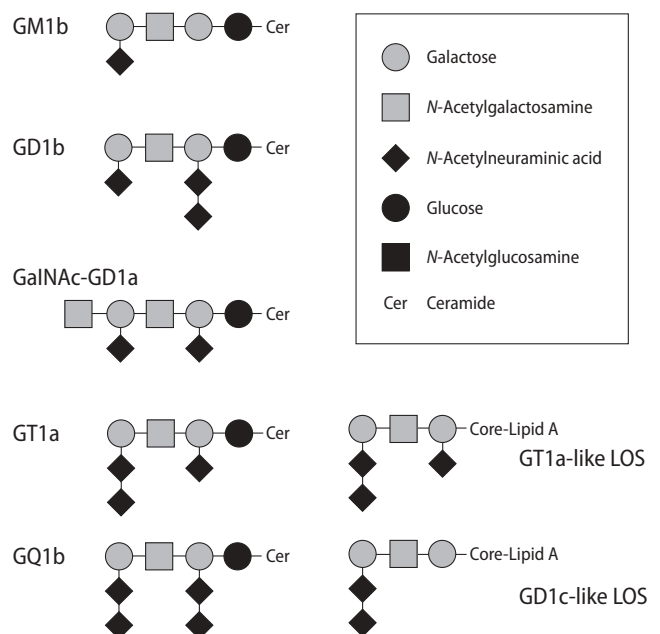


Fig. 1 Structures of GM1b, GD1b, GalNAc-GD1a, GT1a, GQ1b, GT1a-like lipo-oligosaccharide (LOS), and GD1c-like LOS

and sural nerves. H-reflexes were recorded from the soleus muscles after submaximal stimulation of 1-ms duration at the politeal fossa. Somatosensory evoked potentials (SEPs) were recorded by standard procedures after median nerve stimulation at the wrist or tibial nerve stimulation at the ankle. Cervical N13 and cortical N20 were recorded for median nerve stimulation, and lumbar N20 and cortical P37 for tibial nerve stimulation.

Body sway was measured as displacement of the centre of foot pressure detected on a rigid platform mounted on strain gauges using posturography (G5500, Anima Corp., Tokyo, Japan), as described elsewhere [20]. The total length (mm) traveled in 30 seconds was calculated from the centre of foot pressure as subjects stood on the platform. Peak frequency of body sway was calculated by Fourier analysis with a computer program (Gravi Analyzer, Anima Corp.).

Results

■ Patient selection

Based on medical records available to us, 896 patients who showed progressive, relatively symmetric external

ophthalmoplegia and ataxia by four weeks were recruited. Insufficient information for clinical analysis excluded 280 patients and 35 others were excluded because of limb weakness of 3 or less on the MRC scale during the course of illness. The 581 selected patients were divided into three groups: 53 with typical BBE who had impaired consciousness; 466 with typical FS who had alert consciousness and hypo- or areflexia; 62 patients were unclassified because of clear consciousness and preserved muscle stretch reflexes during their illnesses.

■ Clinical profiles

Table 1 shows the clinical profiles of the 53 BBE patients. Males predominated (male/female ratio, 2.3 : 1). The distribution of onset age showed two peaks at 20–29 and 40–49 years (Fig. 2). Ninety-six percent of the patients had antecedent infectious symptoms (52 % upper respiratory infectious symptoms only, 21 % diarrhea only, 8 %

Table 1 Clinical profiles of patients with Bickerstaff's brainstem encephalitis (BBE), Fisher syndrome (FS), and unclassified condition (UC)

	BBE	FS	UC
Number of patients	53	466	62
Sex (male/female)	37/16	281/185	28/34
Age (years): median (range)	40 (0–78)	44 (2–86)	42 (8–88)
Antecedent illness: n (%)			
Upper respiratory infectious symptoms	32 (60 %)	354 (76 %)	42 (67 %)
Diarrhea	15 (29 %)	117 (25 %)	16 (26 %)
Other	9 (16 %)	9 (2 %)	7 (11 %)
Initial symptom: n (%)			
Consciousness disturbance	5 (9 %)	0 (0 %)	0 (0 %)
Diplopia	17 (32 %)	303 (65 %)	38 (61 %)
Blepharoptosis	1 (2 %)	19 (4 %)	1 (2 %)
Limb weakness	0 (0 %)	9 (2 %)	1 (2 %)
Gait disturbance	25 (47 %)	149 (32 %)	15 (24 %)
Dysesthesia	7 (13 %)	65 (14 %)	13 (21 %)
Neurological sign during illness course: n (%)			
Consciousness disturbance	53 (100 %)	0 (0 %)	0 (0 %)
Blepharoptosis	18 (34 %)	172 (37 %)	38 (32 %)
External ophthalmoplegia	53 (100 %)	466 (100 %)	62 (100 %)
Internal ophthalmoplegia	29 (55 %)	163 (35 %)	20 (32 %)
Facial weakness	22 (42 %)	103 (22 %)	10 (16 %)
Bulbar palsy	18 (34 %)	79 (17 %)	5 (8 %)
Mild limb weakness*	26 (49 %)	117 (25 %)	15 (24 %)
Muscle stretch reflex			
Absent or decreased	16 (60 %)	466 (100 %)	0 (0 %)
Normal or brisk	10 (40 %)	0 (0 %)	62 (100 %)
Babinski's sign	20 (38 %)	9 (2 %)	9 (15 %)
Ataxia	53 (100 %)	466 (100 %)	62 (100 %)
Sensory disturbance	22 (42 %)	242 (52 %)	26 (42 %)
Dysesthesia	21 (40 %)	210 (45 %)	25 (40 %)
Superficial sense impairment	8 (15 %)	33 (7 %)	5 (8 %)
Deep sense impairment	1 (2 %)	79 (17 %)	8 (13 %)
Assisted ventilation: n (%)	18 (34 %)	5 (1 %)	0 (0 %)

* ≥ 4 on the Medical Research Council scale

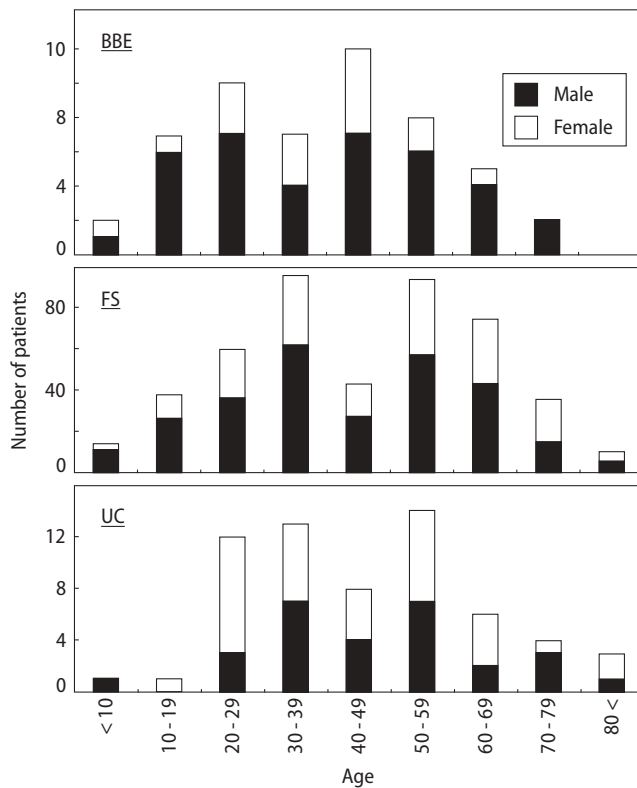


Fig. 2 Age distributions of patients with Bickerstaff's brainstem encephalitis (BBE), Fisher syndrome (FS) and unclassified conditions (UC)

both, 8% fever only, and 8% headache only). The most common initial symptoms were gait disturbance (47%) and diplopia (32%), 9% developing both the same day. According to our strict diagnostic criteria for BBE, all the patients had impaired consciousness (drowsiness 55%, stupor 17%, semicoma 8%, and coma 20%). Blepharoptosis (34%), internal ophthalmoplegia (55%), facial weakness (42%), and bulbar palsy (34%) also were found. Muscle stretch reflexes were absent or decreased in 60%, normal in 13%, and brisk in 27%. Babinski's sign was present in 38%. Deep sense impairment was rare (2%). Median time to nadir was 3 days (range 2–10 days). Treatments were as follows: intravenous immunoglobulin (IVIG) only 35%; combinations of IVIG and methylprednisolone 18%, methylprednisolone only 16%, combinations of plasmapheresis and methylprednisolone 12%, plasmapheresis only 8%, and no specific immunotherapy 11%. Assisted ventilation during the acute phase was required for 34%. Two patients (4%) died; one of aspiration pneumonia 14 days after onset, the other of sudden cardiac arrest 48 days after onset. Patients were admitted for median 35 days (range 8–149 days). At discharge, diplopia persisted in 42%, gait disturbance in 44%, but 13% showed complete remission with no residual symptoms.

Of the 466 FS patients, males predominated (male/fe-

male ratio, 1.5 : 1) (Table 1). Age distribution showed two peaks at 30–39 and 50–59 years (Fig. 2). Ninety percent had had antecedent illnesses; 64% upper respiratory infectious symptoms only, 13% diarrhea only, 12% both, and 2% fever only. The most common initial symptoms were diplopia (65%) and gait disturbance (32%), 14% developing both the same day. All the patients suffered external ophthalmoplegia, ataxia, and hypo- or areflexia. Blepharoptosis (37%), internal ophthalmoplegia (35%), facial weakness (22%), and bulbar palsy (17%) also were present. Extensor plantar response (2%) and need for ventilation support (1%) were rare. Deep sense impairment was infrequent (17%). Median time to nadir was 4 days (range 1–20 days). As treatment, IVIG only was given to 56%, plasmapheresis only to 13%, combined IVIG and methylprednisolone to 7%, plasmapheresis followed by IVIG to 2%, methylprednisolone only to 2%, combined plasmapheresis and methylprednisolone to 2%, combined plasmapheresis and methylprednisolone followed by IVIG to 0.4%, and no specific immunotherapy to 18%. Median hospitalization time was 21 days (range 2–300 days). No patient died. At discharge, diplopia persisted in 71% and gait disturbance in 20%; whereas 16% showed complete remission with no residual symptoms. Table 1 and Fig. 2 give clinical profiles of the 62 unclassified patients.

■ Anti-ganglioside and infectious serology, and CSF study findings

Of the tested anti-ganglioside antibodies, anti-GQ1b IgG antibody was positive in BBE (68%), FS (83%), and the unclassified conditions (58%) (Table 2). Most patients also had anti-GT1a IgG antibody. In contrast, four BBE, four FS, and 10 unclassified patients had anti-GM1b, anti-GalNAc-GD1a, or anti-GD1b IgG antibodies but no anti-GQ1b antibody.

Serological evidence of recent *C. jejuni* (23%) or *H. influenzae* (6%) infection was more common in 34 BBE patients, as well as in FS patients, than in the hospital control participants (Table 3). Anti-*H. influenzae* antibody-positive patients had had symptoms of upper respiratory infections. *C. jejuni* was isolated from one BBE and 16 FS patients. *H. influenzae* was isolated from three BBE and 26 FS patients. In those patients with unclassified conditions, *C. jejuni* infection (18%) commonly preceded onset, whereas *H. influenzae* infection (2%) did not. The *H. influenzae* antibody-positive patient had had symptoms of upper respiratory infections. *C. jejuni* was isolated from one, and *H. influenzae* from three unclassified patients. None of the patients with BBE or unclassified conditions had serological evidence of recent *M. pneumoniae* or cytomegalovirus infection.

During the first week of illness, CSF albuminocytological dissociation occurred in 25% of the BBE,

Table 2 Anti-ganglioside IgG antibodies in patients with Bickerstaff's brainstem encephalitis (BBE), Fisher syndrome (FS), and unclassified conditions (UC)

	BBE	FS	UC
Number of patients	53	466	62
IgG antibody to	n (%)	n (%)	n (%)
GQ1b	36 (68 %)	387 (83 %)	36 (58 %)
GT1a	32 (60 %)	364 (78 %)	41 (66 %)
GQ1b or GT1a	36 (68 %)	401 (86 %)	42 (68 %)
GM1b without GQ1b reactivity	0 (0 %)	9 (2 %)	3 (5 %)
GalNAc-GD1a without GQ1b reactivity	1 (2 %)	9 (2 %)	2 (3 %)
GD1b without GQ1b reactivity	1 (2 %)	9 (2 %)	2 (3 %)
GM1b, GalNAc-GD1a, or GD1b without GQ1b reactivity	2 (4 %)	19 (4 %)	6 (10 %)

Table 3 Infectious serology in patients with Bickerstaff's brainstem encephalitis (BBE), Fisher syndrome (FS), and unclassified conditions (UC)

	BBE	FS	UC	HC	Two-tailed <i>p</i> value		
					BBE vs HC	FS vs HC	UC vs HC
Available data	34	73	44	73			
	n (%)	n (%)	n (%)	n (%)			
<i>Campylobacter jejuni</i>	8 (23 %)	15 (21 %)	8 (18 %)	2 (3 %)	0.0014	0.0007	0.0058
<i>Haemophilus influenzae</i>	2 (6 %)	6 (8 %)	1 (2 %)	0 (0 %)	NS	0.014	NS
Cytomegalovirus	0 (0 %)	3 (4 %)	0 (0 %)	0 (0 %)	NS	NS	NS
<i>Mycoplasma pneumoniae</i>	0 (0 %)	2 (3 %)	0 (0 %)	4 (5 %)	NS	NS	NS

HC hospital controls; NS not significant ($p > 0.05$)

Table 4 CSF findings and abnormal MRI and EEG study findings in patients with Bickerstaff's brainstem encephalitis (BBE), Fisher syndrome (FS), and unclassified condition (UC)

		BBE	FS	UC
CSF	The first week			
	Number of patients	44	375	46
	Albuminocytological dissociation: n (%)	11 (25 %)	139 (37 %)	14 (30 %)
	Cell count (cells/ μ l): median (range)	4 (0–668)	1 (0–105)	1 (0–15)
	Pleocytosis: n (%)	14 (32 %)	15 (4 %)	3 (7 %)
	The second week			
	Number of patients	13	95	13
	Albuminocytological dissociation: n (%)	6 (46 %)	72 (76 %)	7 (54 %)
	Cell count (cells/ μ l): median (range)	3 (0–161)	2 (0–42)	2 (1–61)
	Pleocytosis: n (%)	4 (31 %)	5 (5 %)	2 (15 %)
MRI	Number of patients	47	353	54
	Abnormal findings: n (%)	5 (11 %)	4 (1 %)	4 (7 %)
EEG	Number of patients	30	32	8
	Abnormal findings: n (%)	17 (57 %)	8 (25 %)	0 (0 %)

37 % of the FS and 30 % of the unclassified patients (Table 4): CSF pleocytosis was present in 32 % of the BBE, in 4 % only of the FS and in 7 % of the unclassified patients. In the second week, CSF albuminocytological dissociation occurred in 46 % of the BBE, 76 % of the FS, and 54 % of the unclassified patients. CSF pleocytosis occurred in 31 % of the BBE, in 5 % only of the FS, and in 15 % of the unclassified patients.

■ MRI and neurophysiological findings

Table 4 presents abnormal MRI and EEG findings. On T_2 -weighted images 11 % of 47 BBE patients had high intensity abnormalities in the pons ($n=4$), thalamus ($n=2$), cerebellum ($n=1$), medulla oblongata ($n=1$), midbrain ($n=1$), superior cerebellar peduncle ($n=1$), or corpus callosum ($n=1$). One percent of 353 FS patients had MRI abnormalities in the midbrain ($n=1$), cerebellum ($n=1$), or middle cerebellar peduncle ($n=1$). Seven percent of 54 unclassified patients had abnormalities in the pons ($n=3$), medulla oblongata ($n=2$), or middle cerebellar peduncle ($n=1$). EEG

recordings showed diffuse slow activities in the θ or δ range in 57 % of 30 BBE and 25 % of 32 FS patients.

Table 5 shows results of neurophysiological studies of the subgroup of patients with BBE ($n=4$) or FS ($n=28$). In nerve conduction and H-reflex studies, the most frequent abnormality was the absence of soleus H-reflexes in 75 % of the BBE and 74 % of the FS patients, whereas routine motor and sensory nerve conduction study results were normal for both groups. SEPs were rarely abnormal, and one of the three BBE patients had no cortical potentials (median N20, tibial P37) despite having normal spinal/peripheral potentials (median N13, tibial N20). Similarly, abnormal SEP frequencies were low in FS patients; 23 % had them in median SEPs and 15 % in tibial SEPs. Body sway analyses showed a power spectrum peak at 1-Hz, indicative of proprioceptive afferent system dysfunction in 67 % of the BBE and 72 % of the FS patients. Detailed findings for one BBE patient were reported elsewhere [27].

Discussion

Clinical similarities

The findings of this large study provide significant information about the clinical features of BBE and FS and convincing evidence of their clinical similarity. In both BBE and FS, males predominated, and the age distribution showed two peaks. Symptoms of antecedent infectious illnesses were most common for the upper respiratory system. Blepharoptosis and internal ophthalmoplegia were frequent during the illness. Deep sensation was relatively unimpaired despite profound ataxia.

CSF albuminocytological dissociation occurred in 1/4 of our BBE patients during the first week of illness, increasing to 1/2 in the second week, but each dissociation frequency was lower than in our FS patients. The

presence of CSF albuminocytological dissociation in BBE, as well as in FS, suggests that they have a common etiology. In addition, 1/3 of the BBE patients had CSF pleocytosis, compared to 1/20 of the FS patients. These findings show that CSF studies alone cannot discriminate BBE from FS. Breakdown of the blood-CSF barrier may be more severe in BBE, thereby producing CSF pleocytosis more often than in FS.

Both CNS and PNS involvement

Although the presence of brainstem lesions in BBE was doubted [30], postmortem examinations of three BBE patients found definite inflammatory changes in the brainstem [1, 4, 26]. Our BBE patients also had definite CNS involvement because all had impaired consciousness. Hyperreflexia and Babinski's sign were present in 1/3 of the BBE patients. MRI detected CNS lesions in 1/10 of the BBE patients, and EEG abnormalities in 3/5. In contrast, EEG recordings showed diffuse slow activities in 1/4 of the fully conscious FS patients tested. In the three FS patients, T₂-weighted MRI showed high intensity abnormalities in the brainstem and cerebellum. In an alert FS patient, T₁-weighted MRI showed areas with increased signal intensity in the cerebral white matter, as well as in the brainstem and cerebellum [8]. Supranuclear, nuclear, and internuclear ocular motor signs have been reported in several FS patients [23]. These observations indicate that central components are occasionally associated with FS.

Some investigators have suggested that lesions scattered in the brainstem tegmentum and cerebellar peduncles are responsible for ataxia and areflexia [24], but neurophysiological study results suggest that peripheral abnormalities occur in some FS patients [11, 14, 31]. Similar findings in previous reports include reduced amplitudes of sensory nerve action potentials and the absence of soleus H-reflexes in some FS patients,

Table 5 Nerve conduction study, somatosensory evoked potential, and postural body sway analyses in patients with Bickerstaff's brainstem encephalitis (BBE) and Fisher syndrome (FS)

	BBE		FS	
	Number of examined patients	Abnormality n (%)	Number of examined patients	Abnormality n (%)
Nerve conduction study				
Motor nerve conduction	4	0 (0 %)	28	0 (0 %)
F-wave	4	0 (0 %)	28	0 (0 %)
Sensory nerve conduction	4	0 (0 %)	28	2 (7 %)
Soleus H-reflex*	4	3 (75 %)	23	17 (74 %)
Somatosensory evoked potential				
Median	4	1 (25 %)	13	3 (23 %)
Tibial	3	1 (33 %)	13	2 (15 %)
Postural body sway analysis**	3	2 (67 %)	18	13 (72 %)

* Absent H-reflexes defined as abnormal

** Power spectrum peak at 1-Hz regarded as abnormal

but the number of those examined was too small to reach a definite conclusion. Our study includes 28 patients with FS and 4 with BBE who underwent detailed neurophysiological testing. Results showed that major abnormalities were the absence of soleus H-reflexes and a 1-Hz peak in body sway analyses. Body sway analysis is the classic method used to evaluate the proprioceptive afferent system. Postural sway with a 1-Hz frequency has been reported in patients with tabes dorsalis or ataxic polyneuropathy [3], and a previous FS study found nine of 10 patients had a 1-Hz peak [20]. Our study confirmed these findings and showed that BBE patients also had this peak, indicating that ataxia is caused by a mechanism, dysfunction of the proprioceptive afferent system, common to FS and BBE. Patients with FS or BBE showed both CNS and PNS involvement [8, 27], supporting the hypothesis of continuity between those conditions. The coexistence of central and peripheral components refutes the idea of a simple relationship between BBE and purely central involvement, as well as between FS and a simple peripheral neuropathy.

■ Immunopathogenesis

Most patients with BBE or FS who had been studied had had infectious symptoms before the onset of neurological symptoms. A history of antecedent upper respiratory infection was frequent for both conditions, but most of the causative microorganisms were not identified. Our comprehensive study, however, found *H. influenzae* to be the causative agent in some BBE patients, as reported in FS patients [16]. It is interesting that in Fisher's original publication he reported isolation of *H. influenzae* from the sputum of one of his three patients [10]. Our BBE patients with serologic evidence of recent *H. influenzae* infection all had had upper respiratory infection symptoms. *H. influenzae* was isolated both from those with BBE and with FS, but bacterial isolation is not always indicative of the causative agent, as discussed elsewhere [17]. In both conditions, a history of preceding diarrhea was less common than one of preceding upper respiratory infection symptoms, as reported in GBS [33]. Of previously identified microorganisms, our study showed *C. jejuni* to be the most frequent agent in BBE, as well as in FS [16]. The presence of common, antecedent infectious agents such as *C. jejuni* and *H. influenzae* in both conditions supports the hypothesis that BBE and FS have similar etiologies, as is the case for GBS and FS [34].

Anti-GQ1b IgG antibody was positive in 8/10 of the FS and 7/10 of the BBE patients, but found to be negative in healthy and disease control participants, including those with brainstem infarctions and Wernicke's encephalopathy [7, 35]. The fact that both conditions have this autoantibody in common is strong support for FS and BBE representing a single autoimmune disease.

As stated, *C. jejuni* was the most frequent microbial agent identified in FS and BBE, and was isolated from both groups of patients. Lipo-oligosaccharide (LOS) is a major cell-surface structure expressed by *C. jejuni*. The bacterial sialyltransferase Cst-II functions in the biosynthesis of ganglioside-like LOS, the 51st amino acid determining its enzymatic activity [12]. Cst-II (Thr51) produces GM1-like and GD1a-like LOSs, and *cst-II* (Thr51) strains are associated with GBS [18]. In contrast, Cst-II (Asn51) produces GT1a-like and GD1c-like LOSs (Fig. 1), in which each terminal trisaccharide is identical to that of GQ1b [15, 18]; moreover, *cst-II* (Asn51) strains are associated with FS [18]. We recently showed that the characteristics of a *C. jejuni* isolate from a BBE patient were those of FS not GBS isolates [15]. That isolate had *cst-II* (Asn51) and the GQ1b epitope on LOS, evidence that BBE and FS are part of a continuous spectrum and that BBE is closer to FS than to GBS.

Human immunohistochemical studies showed that GQ1b is highly expressed in extramedullary regions of the ocular motor nerves and some large neurons of dorsal root ganglia [6, 19]. Patients with mild BBE show drowsiness but are easily roused by stimulation, suggesting that the brainstem's reticular formation is involved in BBE. The blood-brain barrier (BBB) protects the brain from deleterious effects of large molecules circulating in the blood, but there are several areas where the BBB is deficient. Studies on the permeability [9] and ultrastructure [32] of the area postrema have demonstrated that the BBB of this region is relatively permeable, allowing large molecules to penetrate into the brainstem parenchyma at this site. It is possible that in BBE the anti-GQ1b antibody reaches the brainstem via this route, and attacks the brainstem reticular formation, but this has yet to be demonstrated experimentally. Anti-GQ1b monoclonal antibody reacts with LOSs from FS-related *H. influenzae* strains [16], indicative that those LOSs have the GQ1b epitope. The possible sequence of events in the pathogenesis of BBE and FS therefore is the following. (1) Infection by a microorganism bearing the GQ1b epitope induces production of anti-GQ1b IgG antibody in certain patients. (2) The anti-GQ1b antibody binds to GQ1b, which is expressed on the oculomotor nerves and primary sensory neurons, inducing FS. (3) In some cases anti-GQ1b antibody also enters the brainstem through areas where the BBB is deficient, e.g. area postrema, and (4) binds to GQ1b which may be expressed on the brainstem reticular formation, thus inducing BBE.

Several patients with FS and ataxic GBS have been reported to have anti-GM1b, anti-GD1b, or anti-GalNAc-GD1a IgG antibody but not anti-GQ1b antibody [25, 36]. Our study adds patients who have either FS or BBE. Although some humans may express GM1b, GD1b, or GalNAc-GD1a in their oculomotor nerves, primary sensory neurons, and brainstems, further immunohistochemical studies are required to test our hypothesis.

■ Proposal of “Fisher-Bickerstaff syndrome” and its treatment

As discussed above, BBE is not distinct from FS clinically, anatomically, or etiologically. These two conditions therefore represent a single autoimmune disease that variably involves the PNS and CNS. Clinical findings, anatomical lesions, and immunological findings for our unclassified condition group were similar to those for BBE and FS. A nosological position is needed for unclassified conditions. FS is familiar to most neurologists, whereas BBE is not. Fisher [10] reported the original cases in 1956, and Bickerstaff and Cloake [5] their first cases in 1951, which they classified as BBE. We propose the new eponymic terminology “Fisher-Bickerstaff syndrome” which includes unclassified conditions as well as FS and BBE.

Randomized controlled trials have established the efficacy of plasma change and IVIG for GBS [13, 29]. Their efficacy for use in Fisher-Bickerstaff syndrome

has yet to be shown because there have been no randomized controlled trials [28]. This retrospective study could also not show whether immunotherapy is effective for Fisher-Bickerstaff syndrome, and whether each therapeutic response differs between FS and BBE. Retrospective analysis of 92 consecutive FS cases, however, showed that IVIG somewhat lessened ophthalmoplegia and ataxia but had no effect on outcomes, presumably because of good natural recoveries [21]. Most of our FS patients had received IVIG or plasmapheresis, which may have been unnecessary. Although BBE outcomes generally are good, several BBE patients have died [1, 4, 26]. We therefore suggest that patients with BBE should be treated with IVIG or plasmapheresis, which should be confirmed in future studies.

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